

HW Ex 2.2 1, 5, 31, 41

HW Ex 2.3 23, 41

- Inverses
- Simple cases
- Equations
- Properties
- How to find it?

§ 2.2 Inverses

Multiplication inverse?

only possible for square matrices.

Def $A: n \times n$ matrix. A is invertible if $\exists$ an $n \times n$ matrix C s.t.
 $AC = CA = I_n$.

In this case, C is the inverse of A .

Unique. C is unique.

Suppose B is also an inverse, then

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

We denote the inverse of A as A^{-1} .

Rank An invertible matrix is also called a nonsingular matrix.

A matrix which is not invertible is singular.

2x2 case

Thm $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

If $ad - bc \neq 0$, then A is invertible and $A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

If $ad - bc = 0$, then A is not inv.

★ Def $ad - bc$ is the determinant of A , denoted as $\det A$

Exmp 2 $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$

$$\det A = 3 \cdot 6 - 4 \cdot 5 = -2 \neq 0, \quad A \text{ inv.}$$

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 6 & -4 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ \frac{5}{2} & -\frac{3}{2} \end{bmatrix}$$

□

One important application towards equations

more important
in theory.

Thm $A: n \times n$ invertible matrix. Then $\vec{b} \in \mathbb{R}^n$,

$$A\vec{x} = \vec{b} \text{ has the unique sol. } \vec{x} = A^{-1}\vec{b}$$

Pf multiply $A\vec{x} = \vec{b}$ by A^{-1} to get $\vec{x} = (A^{-1}A)\vec{x} = A^{-1}\vec{b}$. □

For a real life exmp, see Exmp 3

Exmp 4 Use exmp 2 to solve
$$\begin{cases} 3x_1 + 4x_2 = 3 \\ 5x_1 + 6x_2 = 7 \end{cases}$$

Sol. same as $A\vec{x} = \vec{b} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$

$$\vec{x} = A^{-1}\vec{b} = \begin{bmatrix} -3 & 2 \\ \frac{5}{2} & -\frac{3}{2} \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix} \quad \square$$

Thm 6 A, B invertible $n \times n$ matrices. Then

a) $(A^{-1})^{-1} = A$

→ b) $(AB)^{-1} = B^{-1}A^{-1}$ Caution about the order.

c) $(A^T)^{-1} = (A^{-1})^T$

Pf a) $(A^{-1})A = I$

b) $(AB)(AB)^{-1} = A(BB^{-1})A^{-1} = AA^{-1} = I$

$(AB)^T AB = I$

c) $A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I, \quad (A^{-1})^T A^T = I$. □

Q How to find the inverse?

We need row operations & elementary matrix.

elementary matrix: apply a single elementary row operation on an identity matrix.

Exmp 3 types

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & & \\ & 2 & \\ & & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$E_1 A = \begin{bmatrix} a & b & c \\ d & e & f \\ g-3a & h-3b & i-3c \end{bmatrix}$$

$i=3$

An elementary row operation is the same as left mult. by the corresponding elementary matrix.

The following theorem is the basis of an algorithm to look for an inverse. (a)
Thm A $n \times n$ matrix. A inv. $\Leftrightarrow A$ row equivalent to I_n .
 In this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

Meaning

Algorithm row reduce the augmented matrix $[A \ I]$

If A is row equivalent to I , then $[A \ I]$ is row equivalent to $[I \ A^{-1}]$. ↑ REF

Otherwise, A is not invertible.

Exmp 7

Think about Ex 22. 32